



CCNP Lab: Installing Windows 11 & Software



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Purpose:

The purpose of this lab is to learn how to install windows 11 on any computer. It sets up the fundamental understanding of how a hard drive and flash drive work, and how to install basic software for any computer.

Background:

A **hard drive** is a traditional **data storage** device which is often found inside computers. It helps to store data inside the computer, and even though it is slower and more fragile than modern alternatives, they offer a lot of storage at a low cost. A **flash drive** is a small, portable storage device that uses **flash memory** when plugged in. It's ideal for transferring files or **operating systems**.

Operating systems like Windows serve as the core software which manages a computer's hardware and provides a platform for applications to run. It was first released in 1985, and has gone through several versions, including the most up-to-date version, Windows 11. **Windows 11** is an operating system created by Microsoft in 2021 to allow computers to run with a simple, intuitive design. It's the successor to Windows 10, with new and improved features such as redesigned interfaces, smoother visuals, enhanced performance, and better Microsoft integration of Microsoft apps. To install it from a flash drive you need to install it from a USB via the installation files. By accessing the boot menu, you can choose to access the **BIOS/UEFI settings** and delete the partition which has the existing system and wipe it before installing windows 11.

Rufus is a free tool which creates **bootable USB drives**. By taking an ISO file for Windows or other Operating Systems and writing it to a USB drive, it makes it bootable and able to be installed on other computers and machines. It's especially useful when preparing a flash drive for installing/fixing operating systems.

After installing Windows, a fundamental aspect of installation is **drivers**. They're software programs which **allow the operating system to talk to the hardware components**. Tools such as **Nvidia app** and **Lenovo Vantage** help users update system-specific issues and allow the hardware to have best performance. These companies often update their drivers to **combat bug fixes** and **boost performance**.

Basic productivity and connectivity software also plays a huge role in our modern age. Software such as **Microsoft Teams** allows users to chat, meet via video, and share files. It is often used in workplaces and schools. The base of Teams is called **Office 365**, which is a subscription-based service offered by Microsoft which includes tens of applications including famous ones such as **Word, Excel, PowerPoint, OneDrive, and Excel**. Lastly, **PuTTY** is a **lightweight terminal emulator** which allows

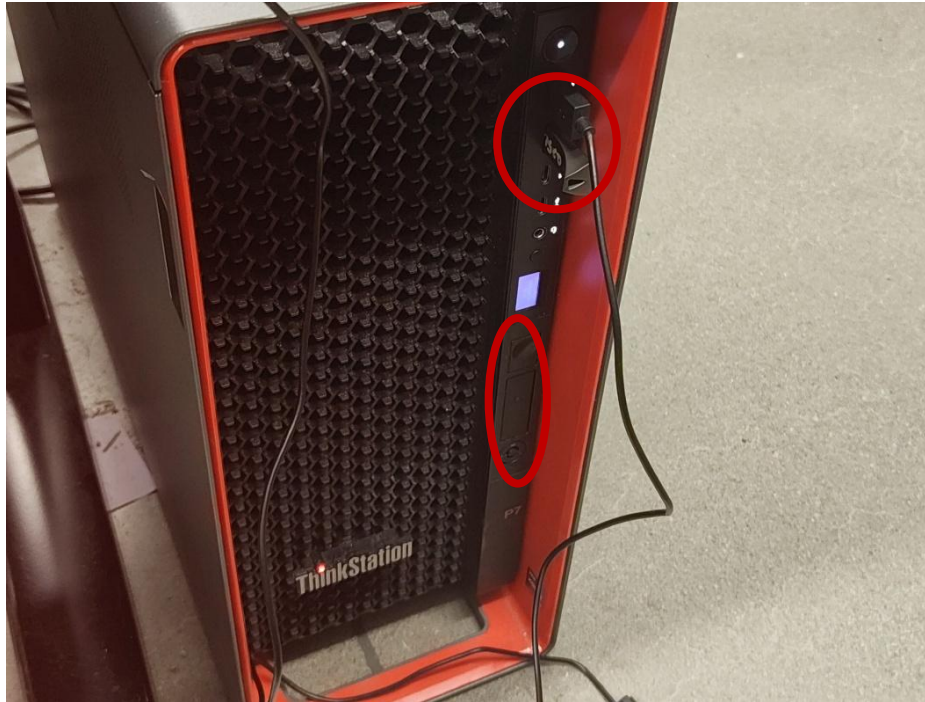
remote servers to connect via **SSH or Telnet**. It's usually used for secure command-line access by developers or IT professionals.

Lab Summary:

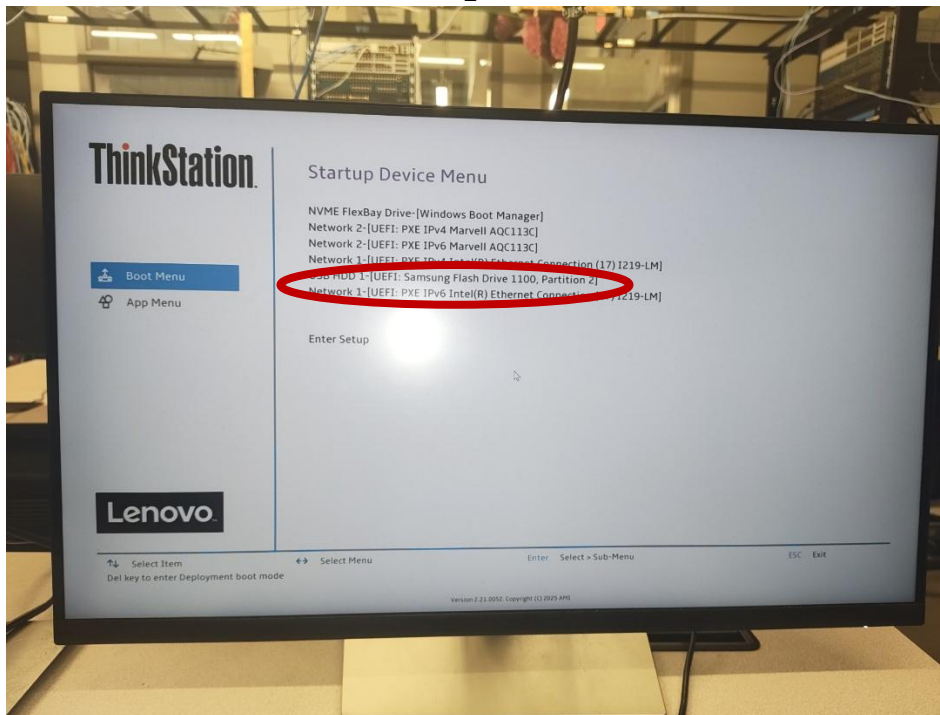
The hard drive I used to install Windows 11 on my machine already had Windows 11 installed on it. As a result, I needed to access the Startup Device Menu by repeatedly clicking F12 when the Lenovo logo popped up in the middle of restarting my device. I selected my flash drive and selected the default settings. I clicked the “Install Windows 11” button and checked off that I agree that everything will be deleted including files, apps, and settings. I deleted everything on Disk 0 except the unallocated space, then selected the unallocated space after I had freed up room to install Windows 11. I went to the Microsoft Store and downloaded an array of useful apps such as PuTTY, Office 365, and Lenovo Vantage. I then downloaded the Nvidia app, Wireshark, and Cisco Packet Tracer by going to their individual websites and installing them. I opened the Nvidia app and Lenovo Vantage and installed all possible drivers by checking my system for any possible updates.

Configs:

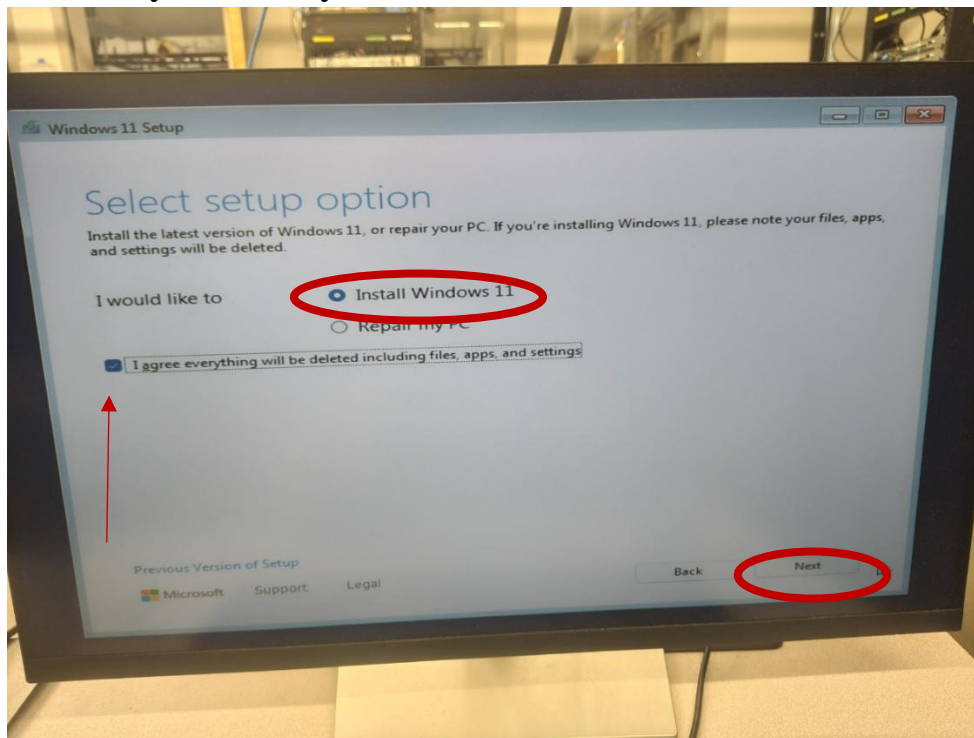
1. Insert hard drive and USB Flash Drive into the PC.



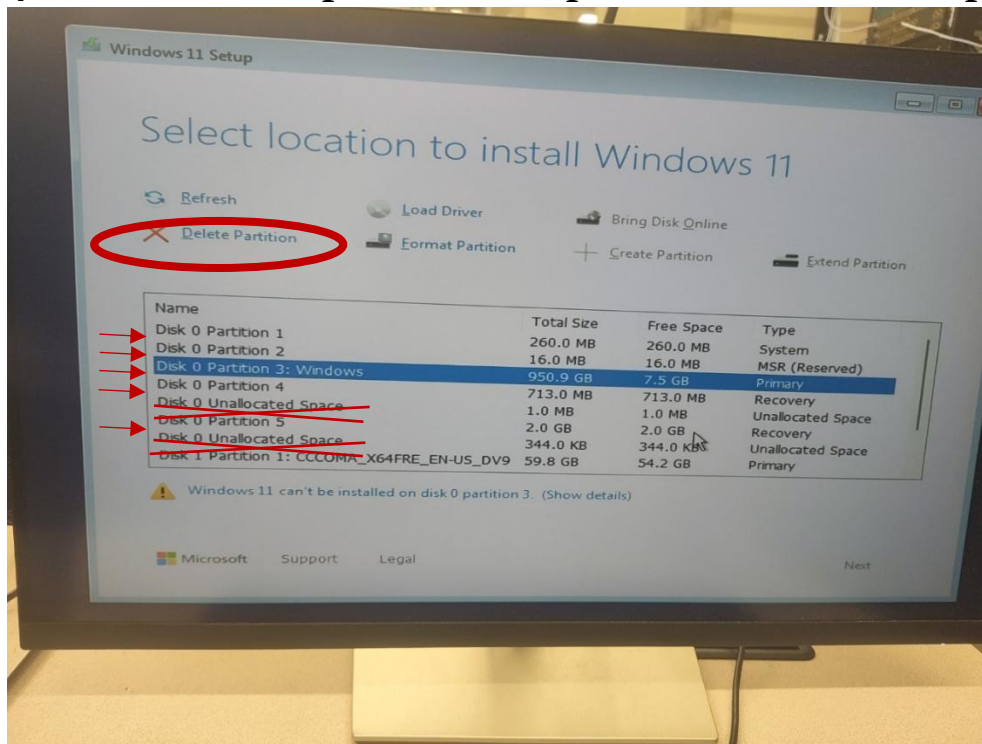
2. Repeatedly press F12 upon loading the PC to access the startup device menu and select the USB as pointed to below.



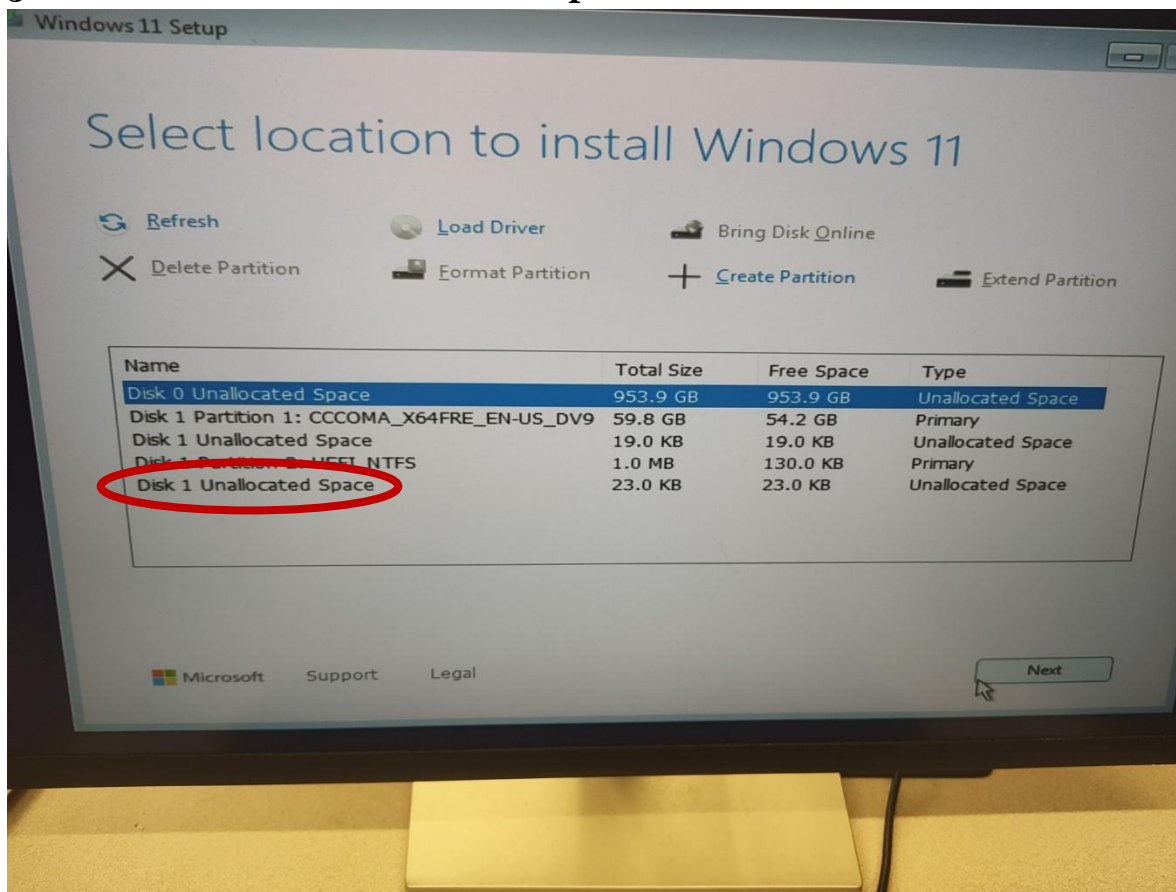
3. Select default language settings, and press “Install Windows 11” and check any necessary boxes to continue.



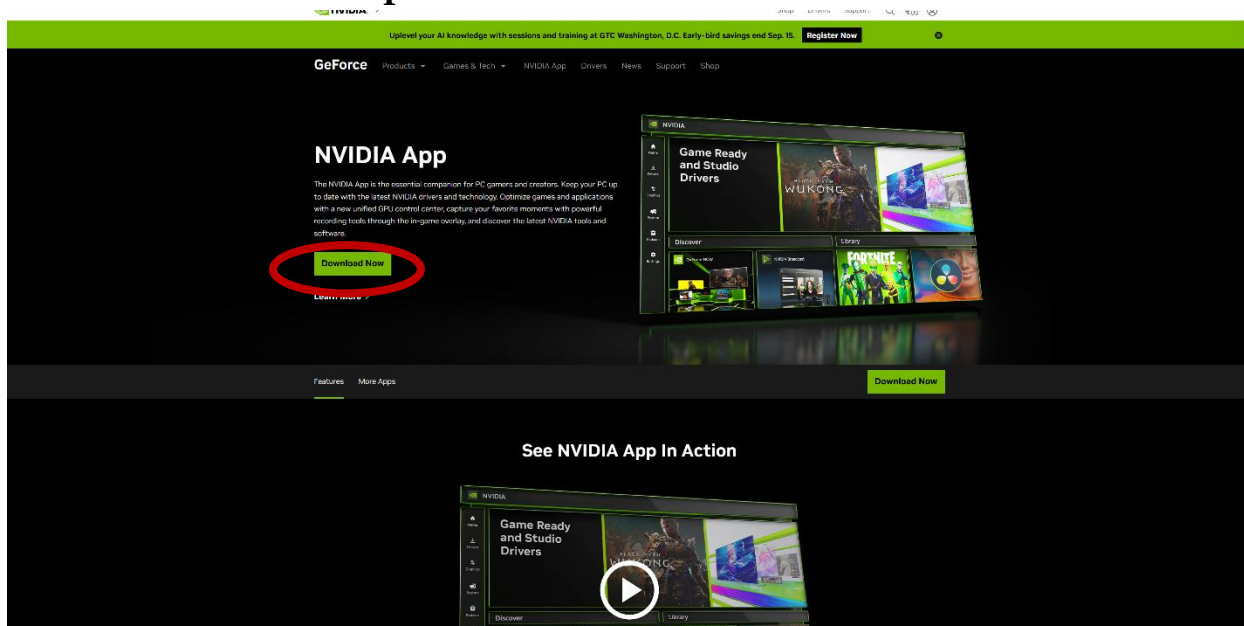
4. Delete all Disk 0 partitions except “Disk 0 Unallocated Space”.



5. Select the “Disk 0 Unallocated Space” and click “Next”.



6. Wait for the computer to load and install drivers or any software needed such as the ones in the pictures below.





Lenovo Commercial Vantage

LENOVO INC.

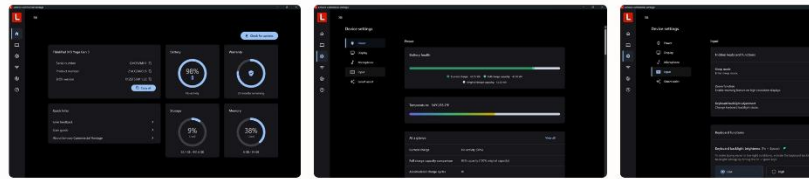
4.3 ★ 190 ratings Productivity

Focused on utility and function, Lenovo Commercial Vantage is an intuitive device management experience that provides access to the hardware features and controls...

Open



Screenshots



Description

Focused on utility and function, Lenovo Commercial Vantage is an intuitive device management experience that provides access to the hardware features and controls - on any Commercial Lenovo Windows 10 PC. Lenovo Commercial Vantage is a free app that enables personal configuration preference and controls for IT admins to control device functionality. Settings and Updates can be facilitated by the user or remotely set by IT. Access and control unique features for features such as energy management, display, camera, audio, keyboard, mouse and pen. **IMPORTANT NOTE: THIS APP IS INTENDED FOR MANAGED ENVIRONMENTS ONLY AND SHOULD NOT BE DOWNLOADED UNLESS ADVISED TO BY YOUR IT ADMINISTRATOR.** Downloading and installing this app without the guidance of an IT administrator could negatively affect the behavior of your device. If you are an end user in an unmanaged environment, please download Lenovo Vantage instead.

Ratings and reviews

4.3
190 RATINGS

Better than consumer version

★★★★★ 5/ MOST HELPFUL FAVORABLE REVIEW

By far the better version of Lenovo Vantage. It not only looks better, it doesn't try to upsell you stuff all the time.

Alex · 3 years ago

190 people found this helpful

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- Stable Release: 4.4.0
- Windows x64 Installer
- Windows x64 PortableApp*
- macOS ARM Disk Image
- macOS Intel Disk Image
- Source Code

Old Stable Release: 4.2.13

Documentation

Latest Release Notes

Wireshark 4.4.9	What's Changed
	wpa-sac-2025-05 SSH dissector crash. Issue 28642.

ALC-CIC
Video Assistant

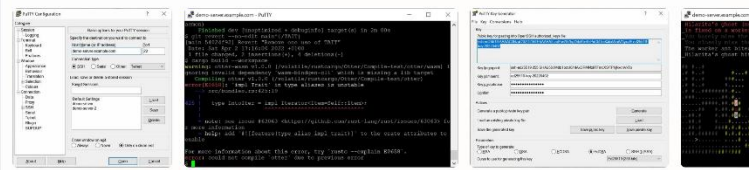
Search

10:42 AM
8/2/2025

Open this product from the Start menu



Screenshots



Description

PuTTY is a communications tool for running interactive command-line sessions on other computers, usually via the SSH protocol. It can also communicate over a serial port, or speak various legacy internet protocols such as Telnet.

Ratings and reviews



Sign in to review

The real PuTTY, from the original author

★★★★★ MOST HELPFUL FAVORABLE REVIEW

Don't bother with the imitations.

Discover more

- Termius - Modern SSH Client
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- WinSCP
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- AbsoluteTelnet SSH Client
Free
- PortIX
Free
- Rufus
Free
- SSH Client - SFTP terminal and Telnet
Free

Problems:

1. Hard Drive containing Windows 11

My hard drive was not clean wiped and contained windows 11 on it, which I identified when I found other things than unallocated space on my hard drive. After some research I figured out that I just needed to delete all partitions related to Disk 0 and proceed as normal.

2. Not being able to login to Office 365

When I attempted to login to Microsoft Teams I found out that I wasn't able to log in. I believe this is a district-related issue or related to the fact I was using the local Antigua account, but I was able to fix this by going to settings and adding a Microsoft work or school account to my machine and then logging in.

3. Not being able to login to the local Antigua account

Upon inserting my hard drive on a different day than the lab to a different machine it wouldn't allow me to enter into the Antigua account without creating a new password and inserting the old one. I managed to bypass this by changing the password and leaving the "Old Password" field empty.

4. Bad Internet

Throughout the lab I consistently had laggy and inconsistent internet. At times I had to reload my Microsoft edge tabs numerous times before they worked and loaded. Despite the CCNP Lab room having 5G Internet, I believe this issue was caused by the excessive number of users on the network and the fact that it was the first day back to the school district which may have caused unforeseen network issues/overload. This problem went away on its own the next day.

Conclusion:

This lab was a very informative one from beginning to end. Even though I had strong knowledge on software and the value behind many of Microsoft's' applications, I learned a lot on how Rufus, hard drives, flash drives, and operating systems work. This knowledge is extremely valuable for basic troubleshooting and for people who want to be able to use machines that haven't already been set up with an Operating System.